

Displays

D1:

$$\bar{\sigma}_{s_0} = \mathcal{F}(s_0; \theta), \text{ where, } \mathcal{F}(s_0; \theta) = \left[\mathbf{F}^1(s_0), \dots, \mathbf{F}^n(s_0), \dots, \mathbf{F}^{(K-1)n+1}(s_0), \dots, \mathbf{F}^{nK}(s_0) \right]^\top \quad (1)$$

D2:

$$R^j = e_\ell^\top s_k - \mathbf{F}^j(s_0), \quad k \in [\mathbf{K}], \ell \in [n]. \quad (2)$$

D3:

$$Y = \left\{ y \mid y = c + \sum_{\ell=1}^m (\mu_\ell v_\ell) \text{ s.t. } P(\mu_1, \dots, \mu_m) = \top \right\}. \quad (3)$$

D4:

$$\ell^* := \lceil (L+1)(1+1/L)(\delta+\tau) \rceil, \quad \ell^* \leq L. \quad (4)$$

D5:

$$\rho := R = \max(\alpha_1 |R^1|, \alpha_2 |R^2|, \dots, \alpha_{nK} |R^{nK}|), \quad (5)$$

D6:

$$\Pr[R < R^*] \geq \delta \iff \Pr[P^*(R^1, \dots, R^{nK}) = \top] \geq \delta, \quad P^*(R^1, \dots, R^{nK}) = \bigwedge_{j=1}^{nK} (|R^j| < \frac{R^*}{\alpha_j}) \quad (6)$$

D7:

$$\delta X = \langle 0_{nK \times 1}, I_{nK}, P^*(R^1, \dots, R^{nK}) \rangle \subset \mathbb{R}^{nK}. \quad (7)$$

D8:

$$\sigma_{s_0}^{\text{sim},q} := s_{t_q+1}, s_{t_q+2}, \dots, s_{t_q+T_i}, \quad t_q = \sum_{\ell=1}^{q-1} T_\ell, \quad t_1 = 0. \quad (8)$$

D9:

$$\text{PE}_i^q = \left[R_i^{t_q n+1}, R_i^{t_q n+2}, \dots, R_i^{(t_q+T_q)n} \right] \quad (9)$$

D10:

$$\overline{\text{PE}}^q = \frac{\sum_{i=1}^{|\mathcal{T}^{\text{trn}}|} \text{PE}_i^q}{|\mathcal{T}^{\text{trn}}|}, \quad \Sigma^q = \frac{\sum_{i=1}^{|\mathcal{T}^{\text{trn}}|} \left(\text{PE}_i^q - \overline{\text{PE}}^q \right)^\top \left(\text{PE}_i^q - \overline{\text{PE}}^q \right)}{|\mathcal{T}^{\text{trn}}|}. \quad (10)$$

D11:

$$\left[r^{t_q n+1}, r^{t_q n+2}, \dots, r^{(t_q+T_q)n} \right] = \mathbf{V}^q \top (\text{PE}^q - \overline{\text{PE}}^q), \quad (11)$$

D12:

$$\rho := \max \left(\frac{|r^1|}{\omega_1}, \frac{|r^2|}{\omega_2}, \dots, \frac{|r^{nK}|}{\omega_{nK}} \right) \quad (12)$$

D13:

$$\omega_j = \max(|r_1^j|, |r_2^j|, \dots, |r_{|\mathcal{T}^{\text{trn}}|}^j|), \quad j \in [nK]. \quad (13)$$

D14:

$$\mathcal{R}^{\text{calib}} = \left\{ (s_{0,i}, \rho_i) \left| \begin{array}{l} s_{0,i} \sim \mathcal{W}, \sigma_{s_{0,i}}^{\text{sim}} \sim \mathcal{D}_{\text{S,K}}^{\text{sim}}, \\ \rho_i = \max(\frac{|r_1^i|}{\omega_1}, \dots, \frac{|r_{nK}^i|}{\omega_{nK}}) \end{array} \right. \right\}. \quad (14)$$

D15:

$$P(r^1, \dots, r^{nK}) = \bigwedge_{q=1}^N P_q(r^{t_q n+1}, \dots, r^{(t_q+T_q)n}), \quad P_q(r^{t_q n+1}, \dots, r^{(t_q+T_q)n}) := \bigwedge_{j=t_q n+1}^{(t_q+T_q)n} (-\omega_j \rho_{\delta,\tau}^* \leq r^j \leq \omega_j \rho_{\delta,\tau}^*), \quad (15)$$

D16:

$$\rho = \max \left(\frac{|r^1|}{\omega_1}, \frac{|r^2|}{\omega_2}, \dots, \frac{|r^{nK}|}{\omega_{nK}} \right) \iff \bigwedge_{j=1}^{nK} [|r^j| \leq \rho \omega_j]. \quad (16)$$

D17:

$$\text{PE}^q = \overline{\text{PE}}^q + \mathbf{V}^q \left[r^{t_q n+1}, r^{t_q n+2}, \dots, r^{(t_q+T_q)n} \right] \quad (17)$$

D18:

$$\mathcal{F}_{10i+j} = (1 - 0.1j)\mathcal{F}_{10i} + 0.1j\mathcal{F}_{10(i+1)} \quad (18)$$

Inline

- I1: $\delta \in (0, 1)$
- I2: δ
- I3: 1
- I4: 2
- I5: $\{1, 2, \dots, n\}$
- I6: $[n]$
- I7: $\oplus$
- I8: $x \sim \mathcal{X}$
- I9: x
- I10: $\mathcal{X}$
- I11: ℓ
- I12: $[n_0, n_1, \dots, n_{\ell+1}]$
- I13: n_0
- I14: $n_{\ell+1}$
- I15: $n_i, i \in [\ell]$
- I16: i
- I17: $e_i \in \mathbb{R}^n$
- I18: $\mathbb{R}^n$
- I19: $\lceil x \rceil$
- I20: $x \in \mathbb{R}$
- I21: $S_0, \dots, S_K \in \mathcal{S}$
- I22: $0, \dots, K$
- I23: $\mathcal{S} \subseteq \mathbb{R}^n$
- I24: $s_1, \dots, s_K$
- I25: $\sigma_{s_0}^{\text{real}}$
- I26: $S_1, \dots, S_K$
- I27: $\mathcal{D}_{S,K}^{\text{real}}$
- I28: S_0
- I29: $\mathcal{W}$
- I30: $\mathcal{I}$
- I31: $\Pr[s_0 \notin \mathcal{I}] = 0$
- I32: $s_0 \sim \mathcal{W}$
- I33: $\sigma_{s_0}^{\text{real}} \sim \mathcal{D}_{S,K}^{\text{real}}$
- I34: $\sigma_{s_0}^{\text{sim}}$
- I35: $\sigma_{s_0}^{\text{sim}} \sim \mathcal{D}_{S,K}^{\text{sim}}$
- I36: $\mathcal{F} : \mathcal{I} \times \Theta \rightarrow \mathcal{S}^K$
- I37: $\theta \in \Theta$
- I38: K
- I39: $\sigma_{s_0}^{\text{sim}} \in \mathcal{S}^K$
- I40: $s_0 \in \mathcal{I}$
- I41: $\mathcal{T}^{\text{trn}}$
- I42: $F^{(k-1)n+\ell}(s_0)$
- I43: ℓ^{th}
- I44: k^{th}
- I45: 3
- I46: $e_\ell \in \mathbb{R}^n$
- I47: $s_1, \dots, s_K$
- I48: $j = (k-1)n + \ell$
- I49: $\rho : \mathbb{R}^{nK} \rightarrow \mathbb{R}_{\geq 0}$
- I50: R^j
- I51: $j \in [nK]$
- I52: $\sigma_{s_0}^{\text{sim}} \sim \mathcal{D}_{S,K}^{\text{sim}}$
- I53: $\rho \sim \mathcal{J}_{S,K}^{\text{sim}}$
- I54: $\mathcal{J}_{S,K}^{\text{sim}}$

I55: $\sigma_{s_0}^{\text{real}} \sim \mathcal{D}_{S,K}^{\text{real}}$
 I56: $\rho \sim \mathcal{J}_{S,K}^{\text{real}}$
 I57: $\mathcal{J}_{S,K}^{\text{real}}$
 I58: $\tau \geq 0$
 I59: $\tau = \text{TV}(\mathcal{J}_{S,K}^{\text{sim}}, \mathcal{J}_{S,K}^{\text{real}})$
 I60: TV
 I61: $\bar{X} \subset \mathbb{R}^{nK}$
 I62: $\mathcal{F}(\mathcal{I}; \theta)$
 I63: $s_0 \in \mathcal{I}$
 I64: $\mathcal{F}(s_0; \theta) \in \bar{X}$
 I65: $Y \subset \mathbb{R}^d$
 I66: $\langle c, V, P \rangle$
 I67: $c \in \mathbb{R}^d$
 I68: $V = \{v_1, v_2, \dots, v_m\}$
 I69: m
 I70: $\mathbb{R}^d$
 I71: $P : \mathbb{R}^m \rightarrow \{\top, \perp\}$
 I72: $d \times m$
 I73: $\mu_\ell \in \mathbb{R}, \ell = 1, \dots, m$
 I74: $\rho_1 < \rho_2 < \dots < \rho_L$
 I75: L
 I76: $\rho_i \sim \mathcal{J}_{S,K}^{\text{sim}}, i \in [L]$
 I77: $\ell := \lceil (L+1)\delta \rceil \leq L$
 I78: $\Pr[\rho < \rho_\ell] \geq \delta$
 I79: ρ_ℓ
 I80: $\text{TV}(\mathcal{J}_{S,K}^{\text{real}}, \mathcal{J}_{S,K}^{\text{sim}}) \leq \tau$
 I81: $\tau > 0$
 I82: $\Pr[\rho < \rho_{\ell^*}] > \delta$
 I83: ρ_{ℓ^*}
 I84: $\text{PE} = [R^1, R^2, \dots, R^{nK}]$
 I85: $\delta \in (0, 1)$
 I86: $\max()$
 I87: $\alpha_j, j \in [nK]$
 I88: α_j
 I89: R
 I90: R^*
 I91: P^*
 I92: $R^j, j = 1, \dots, nK$
 I93: PE
 I94: $-R^*/\alpha_j \leq R^j \leq R^*/\alpha_j$
 I95: $\Pr[P^* = \top] \geq \delta$
 I96: $\delta \in (0, 1)$
 I97: $X \subseteq \mathbb{R}^{nK}$
 I98: $\Pr[\sigma_{s_0}^{\text{real}} \in X] \geq \delta$
 I99: $\sigma_{s_0}^{\text{sim}} \sim \mathcal{D}_{S,K}^{\text{sim}}, s_0 \sim \mathcal{W}$
 I100: $\mathcal{F}(s_0; \theta)$
 I101: $\delta X \subset \mathbb{R}^{nK}$
 I102: $R^j, j \in [nK]$
 I103: $\sigma_{s_0}^{\text{real}} \sim \mathcal{D}_{S,K}^{\text{real}}, s_0 \sim \mathcal{W}$
 I104: $\bar{X}$
 I105: $\mathcal{F}$
 I106: $\text{PE} := [R^1, R^2, \dots, R^{nK}]$
 I107: δX
 I108: $\Pr[\text{PE} \in \delta X] > \delta$
 I109: $X = \bar{X} \oplus \delta X$
 I110: X

I111: $\mathcal{D}_{S,K}^{\text{sim}}$
 I112: $\tau > 0$
 I113: $\mathcal{F}(s_0; \theta)$
 I114: $\mathcal{F}(s_0; \theta)$
 I115: $\sigma_{s_0}^{\text{sim}} := s_1, \dots, s_K$
 I116: N
 I117: T_q
 I118: $q \in [N]$
 I119: $\sigma_{s_0}^{\text{sim},q}, q \in [N]$
 I120: $\mathcal{F}_q(s_0; \theta_q), q \in [N]$
 I121: $\sigma_{s_0}^{\text{sim},q}$
 I122: s_0
 I123: $\bar{X}_q, q \in [N]$
 I124: $\mathcal{F}_q(\mathcal{I}; \theta_q)$
 I125: $\sigma_{s_0}^{\text{sim},q}, q \in [N]$
 I126: $\mathcal{F}_q(s_0; \theta_q), q \in [N]$
 I127: $\bar{X}_q := \mathcal{F}_q(\mathcal{I}; \theta_q)$
 I128: 4
 I129: $\bar{X}_q = \langle \bar{c}_q, \bar{V}^q, \bar{P}_q \rangle$
 I130: $\bar{X} = \langle \bar{c}, \bar{V}, \bar{P} \rangle$
 I131: $\bar{X}_q$
 I132: $\bar{c} = [\bar{c}_1^\top, \dots, \bar{c}_N^\top]^\top$
 I133: $\bar{V} = \mathbf{diag}(\bar{V}^1, \dots, \bar{V}^N)$
 I134: $\bar{P} = \bigwedge_{q=1}^N \bar{P}_q$
 I135: $x_i \in \mathbb{R}^n, i \in [L]$
 I136: $\Sigma \succeq 0, \Sigma \in \mathbb{R}^{n \times n}$
 I137: $x_i, i \in [L]$
 I138: Σ
 I139: $K = 2$
 I140: (R^1, R^2)
 I141: $k = 1$
 I142: (R^3, R^4)
 I143: $k = 2$
 I144: $k = 1, 2$
 I145: (r^1, r^2)
 I146: (r^3, r^4)
 I147: ρ
 I148: $\sigma_{s_0,i}^{\text{sim}} \in \mathcal{T}^{\text{trn}}, i \in [|\mathcal{T}^{\text{trn}}|]$
 I149: $q \in [N]$
 I150: $\sigma_{s_0,i}^{\text{sim},q}$
 I151: $\mathcal{F}_q(s_{0,i}; \theta_q)$
 I152: $q \in [N]$
 I153: $i \in [|\mathcal{T}^{\text{trn}}|]$
 I154: Σ^q
 I155: $\mathbf{V}^q \in \mathbb{R}^{T_q n \times T_q n}$
 I156: $\overline{\text{PE}}^q$
 I157: $\mathbf{V}_\ell^q, \ell \in [T_q n]$
 I158: $\mathbf{V}^q$
 I159: $\text{PE}^q = [R^{t_q n+1}, R^{t_q n+2}, \dots, R^{(t_q+T_q)n}]$
 I160: $r^j, t_q n + 1 \leq j \leq (t_q + T_q)n$
 I161: $\omega_j, j \in [nK]$
 I162: $r_i^j, i \in |\mathcal{T}^{\text{trn}}|$
 I163: $\overline{\text{PE}}^q$
 I164: ω_j
 I165: $\mathcal{R}^{\text{calib}}$
 I166: $\sigma_{s_0,i}^{\text{sim}}, i \in |\mathcal{R}^{\text{calib}}|$

I167: i^{th}
 I168: r_i^j
 I169: $\rho_i \sim \mathcal{J}_{S,K}^{\text{sim}}$
 I170: $\rho_1 < \rho_2 < \dots < \rho_{|\mathcal{R}^{\text{calib}}|}$
 I171: $\text{TV}(\mathcal{J}_{S,K}^{\text{real}}, \mathcal{J}_{S,K}^{\text{sim}}) < \tau$
 I172: ℓ^*
 I173: $\rho_{\delta,\tau}^* := \rho_{\ell^*}$
 I174: $\Pr[\rho < \rho_{\delta,\tau}^*] > \delta$
 I175: $\rho_{\delta,\tau}^*$
 I176: $\text{TV}(\mathcal{J}_{S,K}^{\text{real}}, \mathcal{J}_{S,K}^{\text{sim}}) < \tau$
 I177: $\rho = \max \left(\frac{|r^1|}{\omega_1}, \frac{|r^2|}{\omega_2}, \dots, \frac{|r^{nK}|}{\omega_{nK}} \right)$
 I178: $T_q, q \in [N]$
 I179: $\Pr [P(r^1, \dots, r^{nK}) = \top] > \delta$
 I180: r^j
 I181: $R^j, j \in [nK]$
 I182: $r^j, j \in nK$
 I183: $\Pr[\rho \leq \rho_{\delta,\tau}^*] \geq \delta$
 I184: $\rho < \rho_{\delta,\tau}^* \iff |r^j| < \rho_{\delta,\tau}^* \omega_j$
 I185: $j \in [nK]$
 I186: $\Pr[\bigwedge_{j=1}^{nK} [|r^j| \leq \rho_{\delta,\tau}^* \omega_j]] \geq \delta$
 I187: $T_q, q \in [N]$
 I188: $\square$
 I189: $P_q(r^{t_q n+1}, \dots, r^{(t_q+T_q)n}), q \in [N]$
 I190: $\delta X_q, q \in [N]$
 I191: $\delta X_q = \langle \overline{\text{PE}}^q, \mathbf{V}^q, P_q(r^{t_q n+1}, \dots, r^{(t_q+T_q)n}) \rangle$
 I192: δX_q
 I193: $\delta X = \langle \overline{\text{PE}}, V, P \rangle$
 I194: $V = \mathbf{diag} (V^1, \dots, V^N)$
 I195: $P = \bigwedge_{q=1}^N P_q$
 I196: $X = \bar{X} \oplus \delta X$
 I197: 2
 I198: 5
 I199: $\overline{\text{PE}}^q, q \in [N]$
 I200: $K = 2, n = 2, N = 2$
 I201: $T_1 = T_2 = 1$
 I202: $\sigma_{s_0}^{\text{real}} \in \mathcal{D}_{S,K}^{\text{real}}$
 I203: $\text{TV}(\mathcal{J}_{S,K}^{\text{sim}}, \mathcal{J}_{S,K}^{\text{real}})$
 I204: τ
 I205: $\tau = 0$
 I206: $\tau =$
 I207: $|\mathcal{T}^{\text{trn}}|$
 I208: $|\mathcal{R}^{\text{calib}}|$
 I209: (x_1, x_2, x_3)
 I210: (x_4, x_5, x_6)
 I211: (x_7, x_8, x_9)
 I212: (x_{10}, x_{11}, x_{12})
 I213: $v \sim \mathcal{N}(0_{12 \times 1}, \Sigma_v)$
 I214: $\Sigma_v = \mathbf{diag} \left([0.05 \times \vec{1}_{1 \times 6}, 0.01 \times \vec{1}_{1 \times 6}]^2 \right)$
 I215: $v \sim \mathcal{N}(\vec{0}_{27 \times 1}, \Sigma_v)$
 I216: $\Sigma_v = \mathbf{diag} \left(10^{-5} \times \vec{1}_{1 \times 27} \right)$
 I217: x_{27}
 I218: $\delta t = 0.0005$
 I219: $K = 4000$
 I220: 6

I221: $N = 4000$
 I222: $T_q = 1, q \in [N]$
 I223: $[27, 54, 27]$
 I224: $K = 100$
 I225: $\delta t = 0.05$
 I226: $\delta =$
 I227: $N = 100$
 I228: $T_q = 1$
 I229: $[12, 24, 12]$
 I230: $N = 5000$
 I231: $T_q = 1$
 I232: $i \in 50, 51, \dots, 500$
 I233: $j \in [10]$
 I234: $\mathcal{F}_{10i}$
 I235: Σ_v
 I236: $\tau = 0.04$
 I237: 8
 I238: k
 I239: $\mathcal{D}_{S,K}^{\text{sim}}$
 I240: $\mathcal{D}_{S,K}^{\text{real}}$
 I241: $\mathcal{J}_{S,K}^{\text{sim}}$
 I242: $\mathcal{J}_{S,K}^{\text{real}}$
 I243: V^q
 I244: q
 I245: $\rho \sim \mathcal{J}_{S,K}^{\text{sim}}$
 I246: $\rho \sim \mathcal{J}_{S,K}^{\text{real}}$
 I247: $\delta \in (0, 1)$
 I248: $\Pr[\rho < \rho_\ell] \geq \delta$
 I249: $\Pr[\rho < \rho_{\ell^*}] > \delta$
 I250: $\text{TV}(\mathcal{J}_{S,K}^{\text{real}}, \mathcal{J}_{S,K}^{\text{sim}}) \leq \tau$
 I251: $\Pr[\sigma_{s_0}^{\text{real}} \in X] \geq \delta$
 I252: $\Pr[\text{PE} \in \delta X] > \delta$
 I253: $\Pr[P(r^1, \dots, r^{nK}) = \top] > \delta$
 I254: $\Pr[\rho \leq \rho_{\delta, \tau}^*] \geq \delta$
 I255: $\geq \delta$
 I256: $\rho_{\delta, \tau}^* := \rho_{\ell^*}$
 I257: $\Pr[\cdot]$
 I258: $\tau = 0$
 I259: $s_{t_q + T_i}$
 I260: T_i
 I261: $(\text{PE}^q - \overline{\text{PE}}^q)$
 I262: PE_i^q
 I263: $i \in |\mathcal{T}^{\text{trn}}|$
 I264: $i \in |\mathcal{R}^{\text{calib}}|$
 I265: $j \in nK$
 I266: $\langle \overline{\text{PE}}, V, P \rangle$
 I267: $V = \text{diag}(V^1, \dots, V^N)$
 I268: $s_1, \dots, s_K$
 I269: $S_0, \dots, S_K$
 I270: $\ell^* = \lceil (L+1)(1+1/L)(\delta + \tau) \rceil$
 I271: $\ell^* = 20000 = L$
 I272: $L = 20000$
 I273: $\delta = 0.9999$
 I274: $\ell^* \leq L$
 I275: $\ell^* = 9902 \leq L$
 I276: $L = 10000$

I277: $\delta = 0.95$
I278: $\tau = 0.04$
I279: $i \in 50, \dots, 500$